

PA4 - Sprint Planning

Project: LoveYou – AI-Enhanced Dating Web Application

Duration: 28/07/2026 – 08/08/2026

Sprint Goal: Complete the PA4 requirements by revising the Use-Case Specification, documenting the software architecture, implementing two additional functional groups using Spec Kit, and completing the required development-process evidence and submission package.

1. Sprint Backlog

ID	Product Backlog Item	Priority	Assignee	Estimated Effort
SP4-01	Revise Use-Case Model and Use-Case Specifications based on PA3 feedback	High	Huy	10 h
SP4-02	Create and maintain Changes.md for PA4 revisions	High	Nghĩa	4 h
SP4-03	Create Tech Stack and C4 Level 1 System Context documentation	High	Huy	8 h
SP4-04	Create C4 Level 2 Container Diagram and descriptions	High	Chiến	8 h
SP4-05	Create C4 Level 3 Frontend Component Diagram	High	Huy	8 h
SP4-06	Create C4 Level 3 Backend Component Diagram	High	Chiến	10 h
SP4-07	Create Deployment Diagram and deployment description	High	Nghĩa	6 h
SP4-08	Implement PA4 Functional Group 1 end-to-end using Spec Kit	High	Hoàng + Tấn	20 h
SP4-09	Implement PA4 Functional Group 2 end-to-end using Spec Kit	High	Hoàng + Tấn	20 h
SP4-10	Generate and include Spec Kit test cases	High	Văn	8 h
SP4-11	Integration, bug fixing and architecture consistency review	High	Whole Team	12 h
SP4-12	Prepare AI Usage Report, Weekly Report, Jira evidence and Git log	Medium	Nghĩa	8 h

ID	Product Backlog Item	Priority	Assignee	Estimated Effort
SP4-13	Prepare narrated demo video and final PA4 submission package	High	Whole Team	8 h

2. Sprint Tasks

2.1 Requirements Revision

Task	Owner	Expected Output
Review PA3 TA feedback	Tường Huy	Feedback checklist
Revise Use-Case Model	Tường Huy	Revised Use-Case Model
Revise Use-Case Specifications	Tường Huy	Revised Use-Case Specifications
Record all PA3-to-PA4 changes	Nghĩa	Changes.md
Cross-check requirements against implemented features	Whole Team	Consistency review

2.2 Software Architecture

Task	Owner	Expected Output
Review actual source-code architecture	Chiến + Huy	Architecture baseline
Document Tech Stack	Huy	Tech Stack section
Create C4 Level 1 System Context	Huy	System Context Diagram
Create C4 Level 2 Container Diagram	Chiến	Container Diagram
Create C4 Level 3 Frontend Component Diagram	Huy	Frontend Component Diagram
Create C4 Level 3 Backend Component Diagram	Chiến	Backend Component Diagram
Create Deployment Diagram	Nghĩa	Deployment Diagram
Verify architecture against source code	Whole Team	Architecture consistency checklist

2.3 PA4 Feature Development

Task	Owner	Expected Output
Select and confirm Functional Group 1	Whole Team	Confirmed PA4 FG
Select and confirm Functional Group 2	Whole Team	Confirmed PA4 FG
Create Spec Kit specification for FG1	Hoàng + Tấn	specs/ artifacts
Create Spec Kit plan/tasks for FG1	Hoàng + Tấn	plans/ , tasks/ artifacts

Task	Owner	Expected Output
Implement FG1 frontend/backend/database flow	Hoàng + Tấn	Working FG1
Create Spec Kit specification for FG2	Hoàng + Tấn	specs/ artifacts
Create Spec Kit plan/tasks for FG2	Hoàng + Tấn	plans/, tasks/ artifacts
Implement FG2 frontend/backend/database flow	Hoàng + Tấn	Working FG2
Generate required test cases	Trọng Văn	Test artifacts

2.4 Integration and Verification

Task	Owner	Expected Output
Integrate frontend and backend	Whole Team	Integrated Build
Verify database operations	Assigned developers	Working persistence
Run generated test cases	Trọng Văn + developers	Test results
Fix critical integration defects	Whole Team	Stable PA4 Build
Check architecture-documentation consistency	Whole Team	Final architecture review

2.5 Submission and Process Evidence

Task	Owner	Expected Output
Prepare AI Usage Report	Nghĩa	AI Usage Report
Prepare Weekly Report	Nghĩa	Weekly Report
Capture Jira task assignment/progress	Nghĩa	Jira screenshots
Export Git log	Nghĩa	Git log evidence
Record narrated demo	Whole Team	YouTube video
Convert Markdown documents to PDF	Whole Team	PDF package
Final submission review	Whole Team	PA4 ZIP

3. Sprint Schedule

Date	Activities
28/07	Sprint Planning and PA4 task assignment
29/07 - 31/07	Use-Case revision and architecture baseline
30/07	Scrum Meeting 1
01/08 - 04/08	C4 Level 1/2/3 and Deployment documentation

Date	Activities
02/08 - 06/08	Spec Kit workflow and implementation of two functional groups
05/08	Scrum Meeting 2
06/08 - 07/08	Integration testing, bug fixing and architecture consistency review
08/08	Sprint Review, Sprint Retrospective and PA4 submission

4. Definition of Done

A backlog item is considered completed when:

- The required implementation or document is completed.
- Source code and related artifacts are committed to GitHub.
- Documentation has been reviewed by another team member.
- Implemented functional groups work end-to-end across frontend, backend/API and database where applicable.
- Required Spec Kit artifacts are included.
- Generated test cases are included.
- No critical integration defects remain.
- Architecture documentation is consistent with the actual implementation.
- Required Markdown and PDF versions are prepared.
- The Product Owner/team accepts the deliverable.

5. Risks

Risk	Impact	Mitigation
Architecture documentation does not match source code	High	Review diagrams against actual routes, components, APIs and database structure before submission
Spec Kit artifacts are incomplete	High	Follow the full specification → plan → tasks → implementation workflow and verify the artifact folders
Two functional groups cannot be completed end-to-end	High	Confirm scope early and prioritize critical frontend/backend/database paths
Integration conflicts	Medium	Frequent commits, pull requests and code reviews
Generated tests fail after implementation	Medium	Run tests during integration and fix critical failures before submission
Documentation takes time away from implementation	Medium	Assign documentation owners and review continuously
Missing submission evidence	High	Maintain a PA4 checklist for Jira, Git log, AI report, weekly report and demo video

Risk	Impact	Mitigation
Schedule delay	High	Prioritize Section E implementation and required architecture artifacts first

6. Sprint Ceremonies

Event	Schedule
Sprint Planning	28/07/2026
Daily Scrum	Regular short team meetings
Scrum Meeting 1	30/07/2026
Scrum Meeting 2	05/08/2026
Sprint Retrospective	08/08/2026

7. Expected Deliverables

- Revised Use-Case Model
- Revised Use-Case Specifications
- **Changes.md**
- Tech Stack documentation
- C4 Level 1 System Context Diagram
- C4 Level 2 Container Diagram
- C4 Level 3 Frontend Component Diagram
- C4 Level 3 Backend Component Diagram
- Deployment Diagram
- Two implemented PA4 Functional Groups
- Spec Kit specifications, plans and tasks
- Generated test cases
- Complete source code excluding generated/dependency directories
- Narrated demo video with YouTube link
- AI Usage Report
- Weekly Report
- Jira screenshots
- Git log
- Markdown and PDF documentation
- Final PA4 submission package

8. Acceptance Criteria

Sprint 4 is considered successful if:

- Revised requirements are consistent with PA3 feedback and current implementation.

- `Changes.md` clearly records changes from PA3.
- C4 Level 1, Level 2 and Level 3 architecture diagrams are complete and consistent with the source code.
- Deployment architecture is documented using Mermaid.
- Two additional functional groups are implemented end-to-end using the required Spec Kit workflow.
- Spec Kit specifications, plans, tasks and generated test cases are included.
- The implemented features can be demonstrated in the narrated video.
- AI usage, weekly process, Jira progress and Git history are documented.
- Required Markdown and PDF files are complete.
- The final PA4 submission package is reviewed by all team members before submission.